

STAR TOPOLOGY

1. Project Overview

This document explicitly highlights the design, configuration and implementation of the local area network using a logical star topology that is built using the Packet Tracer software. In this project the principle of the star is shown, and all devices are placed on different VLANs within different departments, communication is greatly facilitated by the multilayered switch using IP routing.

- Project Title: Star_PathFinder
- Objective: The aim of the project is to demonstrate how data flows from a single point in a network setup where all the devices gather their data from.
- Topology: Star Topology
- Core technology: VLAN with segmentation

2. Materials used

Item	Quantity	Model
PC	10	PC-PT
Access Switches	5	2960-24TT
Multilayered switch	1	3560-24PS

3. Network design and Topology

Physical topology (hierarchical star)

- Core switch: One multilayered switch that handles all the routing
- Access/distribution switches: 5 connected directly to the core.
- End devices: 10 PCs connected to the distribution switches.

Logical design with VLAN segmentation:

Each access switch is able to handle all the multiple VLAN in the system due to communication being enabled by the multilayered switch via ip routing.

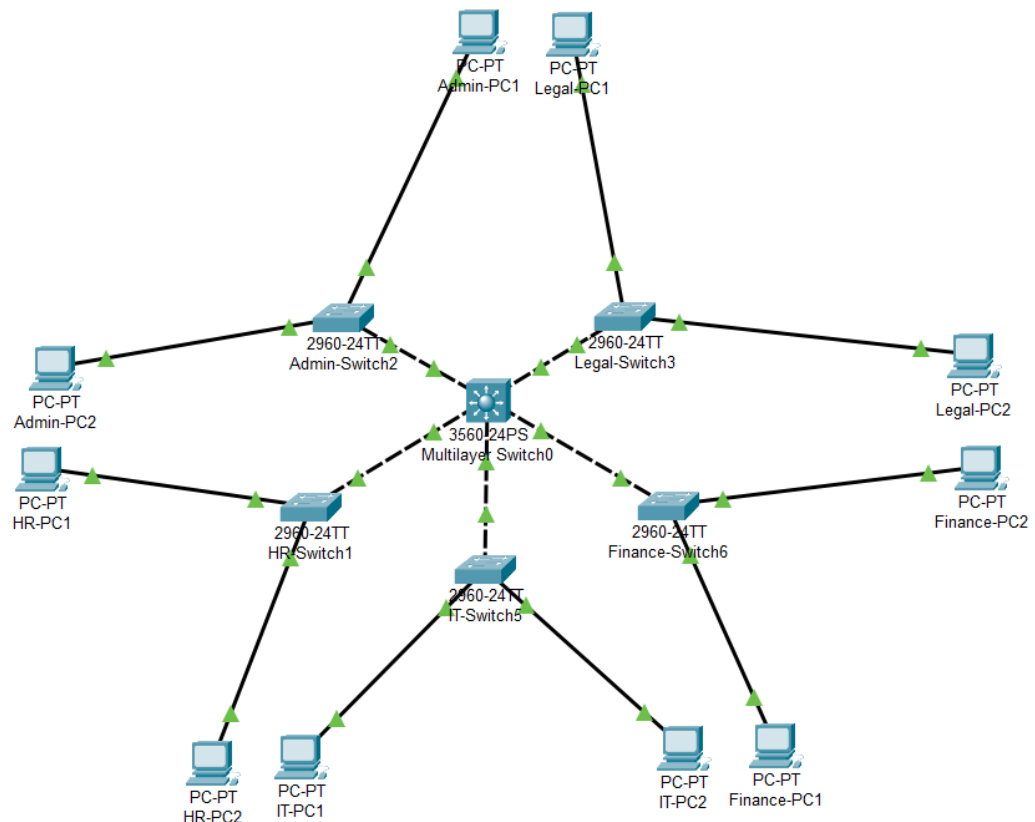


Diagram explanation:

The multilayered switch acts as a core for the system with all the departmental switches connected to it. The aim of doing such is for the layered switch to act as a router in order to ensure that devices in different and same departments are able to send communication as required and whenever possible. The department switches are not connected to each other but directly to the switch so that it acts as a single point of data sending and receiving. The access switches further have two end devices connected to them. The information is meant to flow from the PCs to the access switches of the relevant department then to the multilayered switch if it has the message is sent to a different department, otherwise the departmental access switch takes care of it.

4. Configuration

- The core switch (Multilayered switch)
! Basic Configuration

```
Core-MLS> enable
Core-MLS# configure terminal
Core-MLS(config)# hostname Core-MLS
```

! Create VLANs according to departmental scheme

```
Core-MLS(config)# vlan 10
Core-MLS(config-vlan)# name ADMIN
Core-MLS(config-vlan)# exit
Core-MLS(config)# vlan 20
Core-MLS(config-vlan)# name HR
Core-MLS(config-vlan)# exit
Core-MLS(config)# vlan 30
Core-MLS(config-vlan)# name IT
Core-MLS(config-vlan)# exit
Core-MLS(config)# vlan 40
Core-MLS(config-vlan)# name FINANCE
Core-MLS(config-vlan)# exit
Core-MLS(config)# vlan 50
Core-MLS(config-vlan)# name LEGAL
Core-MLS(config-vlan)# exit
```

! Configure trunk ports to access switches (Fa0/1-5)

```
Core-MLS(config)# interface range fastEthernet 0/1-5
Core-MLS(config-if-range)# switchport trunk encapsulation
dot1q
Core-MLS(config-if-range)# switchport mode trunk
Core-MLS(config-if-range)# switchport trunk native vlan 999
Core-MLS(config-if-range)# no shutdown
Core-MLS(config-if-range)# exit
```

! Configure SVIs for inter-VLAN routing

```
Core-MLS(config)# interface vlan 10
Core-MLS(config-if)# description ADMIN_DEPARTMENT
Core-MLS(config-if)# ip address 192.168.10.1 255.255.255.0
Core-MLS(config-if)# no shutdown
```

```
Core-MLS(config-if)# exit
```

```
Core-MLS(config)# interface vlan 20
```

```
Core-MLS(config-if)# description HR_DEPARTMENT
```

```
Core-MLS(config-if)# ip address 192.168.20.1 255.255.255.0
```

```
Core-MLS(config-if)# no shutdown
```

```
Core-MLS(config-if)# exit
```

```
Core-MLS(config)# interface vlan 30
```

```
Core-MLS(config-if)# description IT_DEPARTMENT
```

```
Core-MLS(config-if)# ip address 192.168.30.1 255.255.255.0
```

```
Core-MLS(config-if)# no shutdown
```

```
Core-MLS(config-if)# exit
```

```
Core-MLS(config)# interface vlan 40
```

```
Core-MLS(config-if)# description FINANCE_DEPARTMENT
```

```
Core-MLS(config-if)# ip address 192.168.40.1 255.255.255.0
```

```
Core-MLS(config-if)# no shutdown
```

```
Core-MLS(config-if)# exit
```

```
Core-MLS(config)# interface vlan 50
```

```
Core-MLS(config-if)# description LEGAL_DEPARTMENT
```

```
Core-MLS(config-if)# ip address 192.168.50.1 255.255.255.0
```

```
Core-MLS(config-if)# no shutdown
```

```
Core-MLS(config-if)# exit
```

```
! Enable IP routing
```

```
Core-MLS(config)# ip routing
```

```
! Save configuration
```

```
Core-MLS(config)# exit
```

```
Core-MLS# copy running-config startup-config
```

- The access switches (departmental switches)

```
ADMIN DEPARTMENT
```

```
! Basic Configuration
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-ADMIN
```

```
! Create VLAN 10
SW-ADMIN(config)# vlan 10
SW-ADMIN(config-vlan)# name ADMIN
SW-ADMIN(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
SW-ADMIN(config)# interface range fastEthernet 0/1-2
SW-ADMIN(config-if-range)# switchport mode access
SW-ADMIN(config-if-range)# switchport access vlan 10
SW-ADMIN(config-if-range)# spanning-tree portfast
SW-ADMIN(config-if-range)# no shutdown
SW-ADMIN(config-if-range)# exit
```

```
! Configure trunk port to Core MLS (Fa0/3)
SW-ADMIN(config)# interface fastEthernet 0/3
SW-ADMIN(config-if)# switchport mode trunk
SW-ADMIN(config-if)# switchport trunk native vlan 999
SW-ADMIN(config-if)# no shutdown
SW-ADMIN(config-if)# exit
```

```
! Save configuration
SW-ADMIN(config)# exit
SW-ADMIN# copy running-config startup-config
```

HR DEPARTMENT :

```
! Basic Configuration

Switch> enable

Switch# configure terminal
```

```
Switch(config)# hostname SW-HR
```

```
! Create VLAN 20
```

```
SW-HR(config)# vlan 20
```

```
SW-HR(config-vlan)# name HR
```

```
SW-HR(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
```

```
SW-HR(config)# interface range fastEthernet 0/1-2
```

```
SW-HR(config-if-range)# switchport mode access
```

```
SW-HR(config-if-range)# switchport access vlan 20
```

```
SW-HR(config-if-range)# spanning-tree portfast
```

```
SW-HR(config-if-range)# no shutdown
```

```
SW-HR(config-if-range)# exit
```

```
! Configure trunk port to Core MLS (Fa0/3)
```

```
SW-HR(config)# interface fastEthernet 0/3
```

```
SW-HR(config-if)# switchport mode trunk
```

```
SW-HR(config-if)# switchport trunk native vlan 999
```

```
SW-HR(config-if)# no shutdown
```

```
SW-HR(config-if)# exit
```

```
! Save configuration
```

```
SW-HR(config)# exit
```

```
SW-HR# copy running-config startup-config
```

IT DEPARTMENT:

! Basic Configuration

Switch> enable

Switch# configure terminal

Switch(config)# hostname SW-IT

! Create VLAN 30

SW-IT(config)# vlan 30

SW-IT(config-vlan)# name IT

SW-IT(config-vlan)# exit

! Configure access ports for PCs (Fa0/1-2)

SW-IT(config)# interface range fastEthernet 0/1-2

SW-IT(config-if-range)# switchport mode access

SW-IT(config-if-range)# switchport access vlan 30

SW-IT(config-if-range)# spanning-tree portfast

SW-IT(config-if-range)# no shutdown

SW-IT(config-if-range)# exit

! Configure trunk port to Core MLS (Fa0/3)

SW-IT(config)# interface fastEthernet 0/3

SW-IT(config-if)# switchport mode trunk

SW-IT(config-if)# switchport trunk native vlan 999

SW-IT(config-if)# no shutdown

SW-IT(config-if)# exit

! Save configuration

SW-IT(config)# exit

SW-IT# copy running-config startup-config

FINANCE DEPARTMENT:

! Basic Configuration

Switch> enable

Switch# configure terminal

Switch(config)# hostname SW-FINANCE

! Create VLAN 40

SW-FINANCE(config)# vlan 40

SW-FINANCE(config-vlan)# name FINANCE

SW-FINANCE(config-vlan)# exit

! Configure access ports for PCs (Fa0/1-2)

SW-FINANCE(config)# interface range fastEthernet 0/1-2

SW-FINANCE(config-if-range)# switchport mode access

SW-FINANCE(config-if-range)# switchport access vlan 40

SW-FINANCE(config-if-range)# spanning-tree portfast

SW-FINANCE(config-if-range)# no shutdown

SW-FINANCE(config-if-range)# exit

! Configure trunk port to Core MLS (Fa0/3)


```
SW-FINANCE(config)# interface fastEthernet 0/3
SW-FINANCE(config-if)# switchport mode trunk
SW-FINANCE(config-if)# switchport trunk native vlan 999
SW-FINANCE(config-if)# no shutdown
SW-FINANCE(config-if)# exit

! Save configuration
SW-FINANCE(config)# exit
SW-FINANCE# copy running-config startup-config
```

LEGAL DEPARTMENT

```
! Basic Configuration

Switch> enable

Switch# configure terminal

Switch(config)# hostname SW-LEGAL

! Create VLAN 50

SW-LEGAL(config)# vlan 50
SW-LEGAL(config-vlan)# name LEGAL
SW-LEGAL(config-vlan)# exit

! Configure access ports for PCs (Fa0/1-2)

SW-LEGAL(config)# interface range fastEthernet 0/1-2
SW-LEGAL(config-if-range)# switchport mode access
SW-LEGAL(config-if-range)# switchport access vlan 50
```

```
SW-LEGAL(config-if-range)# spanning-tree portfast
```

```
SW-LEGAL(config-if-range)# no shutdown
```

```
SW-LEGAL(config-if-range)# exit
```

! Configure trunk port to Core MLS (Fa0/3)

```
SW-LEGAL(config)# interface fastEthernet 0/3
```

```
SW-LEGAL(config-if)# switchport mode trunk
```

```
SW-LEGAL(config-if)# switchport trunk native vlan 999
```

```
SW-LEGAL(config-if)# no shutdown
```

```
SW-LEGAL(config-if)# exit
```

! Save configuration

```
SW-LEGAL(config)# exit
```

```
SW-LEGAL# copy running-config startup-config
```

- The end devices PCs

5. IP Addressing

The devices in the network are placed on different VLANs depending on which department they belong to, the addresses are static and manually given using IPv4 only.

- VLAN 10 – Admin Department
 - ◆ Network address: 192.168.10.0/24
 - ◆ Subnet Mask: 255.255.255.0
 - ◆ Default gateway: 192.168.10.1
- VLAN 20- HR Department
 - Network Address:192.168.20.0/24

- Subnet mask:255.255.255.0
- Default gateway: 192.168.20.1

- VLAN 30- IT Department
 - Network Address: 192.168.30.0/24
 - Subnet mask: 255.255.255.0
 - Default gateway: 192.168.30.1

- VLAN 40- Finance Department
 - Network Address: 192.168.40.0/24
 - Subnet mask: 255.255.255.0
 - Default gateway: 192.168.40.1

- VLAN 50- Legal Department
 - Network Address: 192.168.50.0/24
 - Subnet Mask: 255.255.255.0
 - Default gateway: 192.168.50.1

All devices on this setup are wired.

Device	IP address	VLAN	Remark
Admin-PC1	192.168.10.2	10	Connected to the admin switch
Admin-PC2	192.168.10.3	10	Connected to admin switch
HR-PC1	192.168.20.2	20	Connected to the HR switch
HR-PC2	192.168.20.3	20	Connected to HR switch
IT-PC1	192.168.30.2	30	Connected to IT switch
IT-PC2	192.168.30.3	30	Connected to IT switch
Finance-PC1		40	Connected to the Finance switch
Finance-PC2		40	Connected to the finance switch
Legal-PC1	192.168.50.3	50	Connected to the legal switch

Legal-PC2	192.168.50.2	50	Connected to legal switch
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6. Testing and verification

a)The cdp neighbors as seen by the core switch

```
Switch#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch           Fas 0/1           163        S            2960        Fas 0/3
Switch           Fas 0/5           163        S            2960        Fas 0/3
Switch           Fas 0/3           163        S            2960        Fas 0/3
Switch           Fas 0/4           163        S            2960        Fas 0/3
Switch           Fas 0/2           163        S            2960        Fas 0/3
Switch#
```

b)The core switch is able to ping the default gateways of each department, thus confirming that there is connection established between the multilayered switch and the departmental switches.

```
Switch#ping 192.168.10.1
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/8 ms
```

```
Switch#ping 192.168.20.1
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/6 ms
```

```
Switch#ping 192.168.30.1
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
```

```
Switch#ping 192.168.40.1
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/6 ms
```

```
Switch#ping 192.168.50.1
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/5 ms
```

c) This snippet indicates that the core switch is able to communicate effectively with the end devices and get a reply

```
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Switch#ping 192.168.30.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Switch#ping 192.168.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

Switch#ping 192.168.50.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

Switch#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

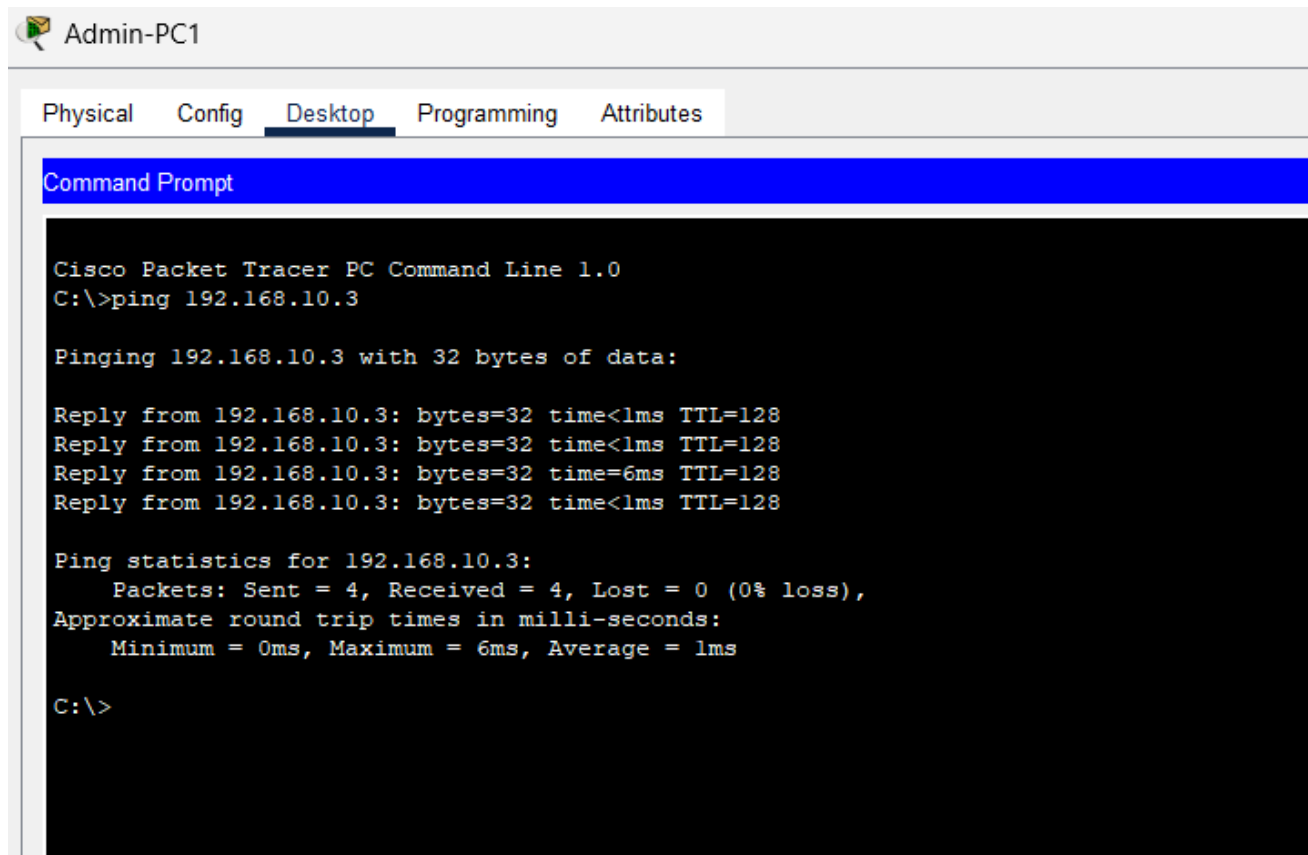
Switch#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

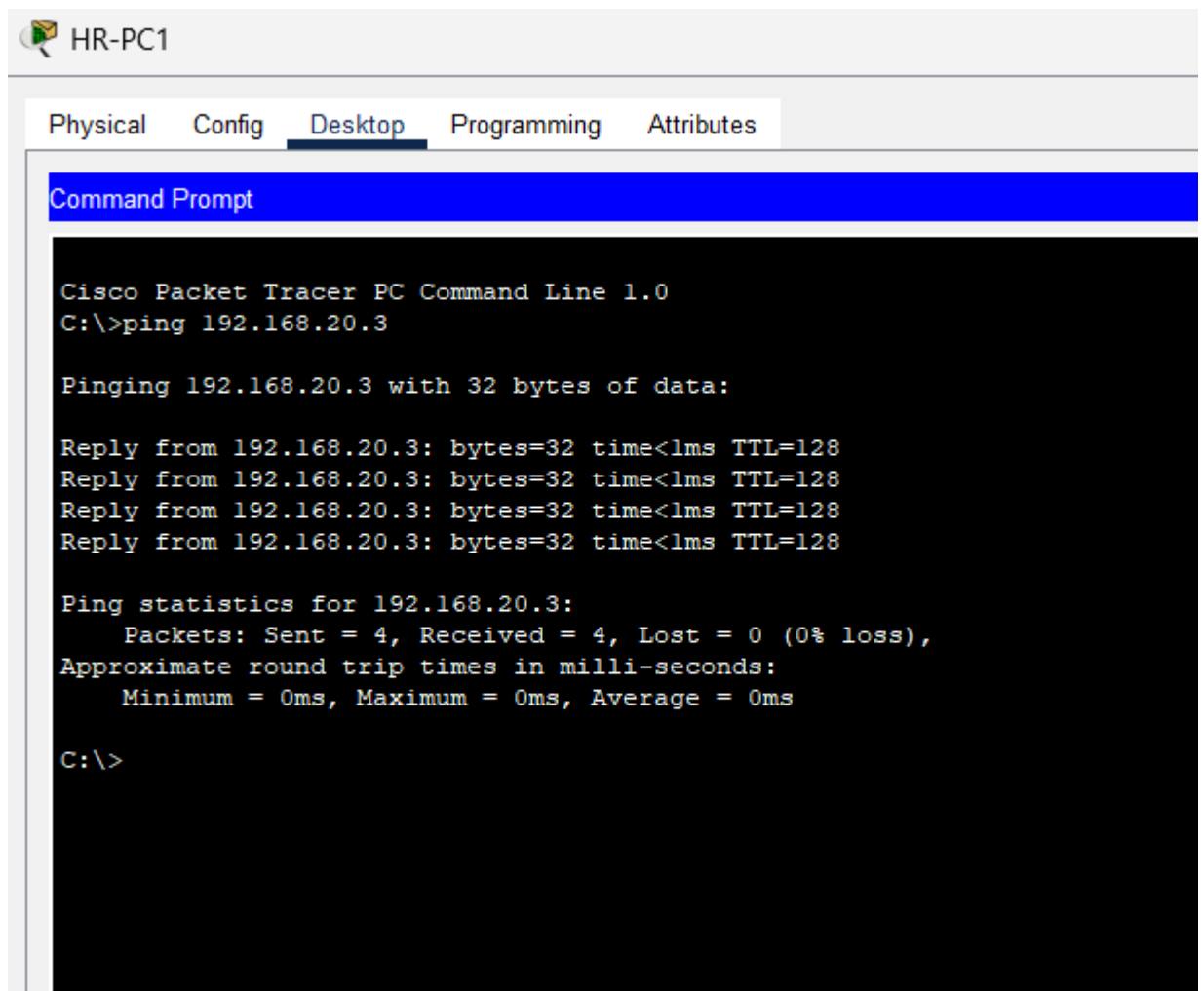
Switch#
```

Communication within the same departments

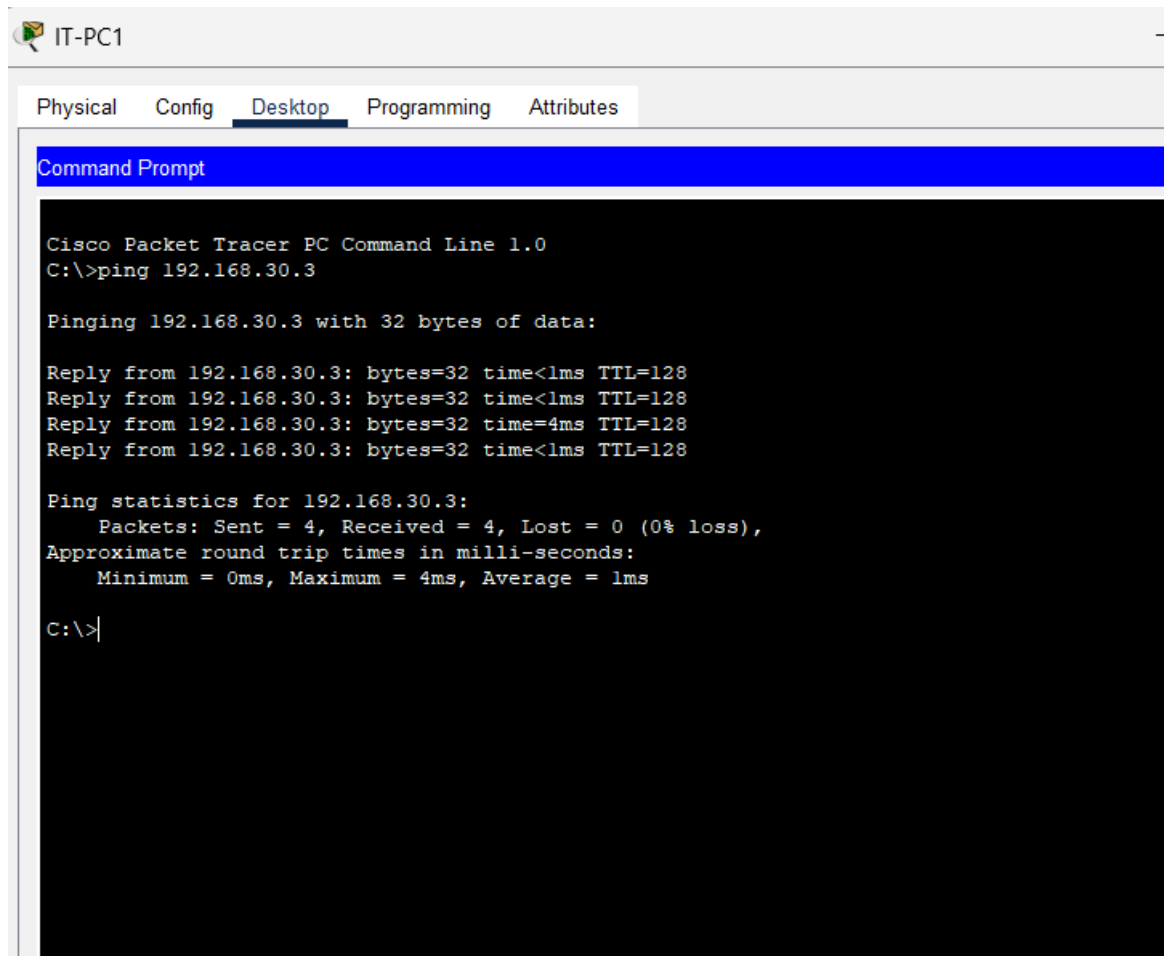
d. Admin dept:



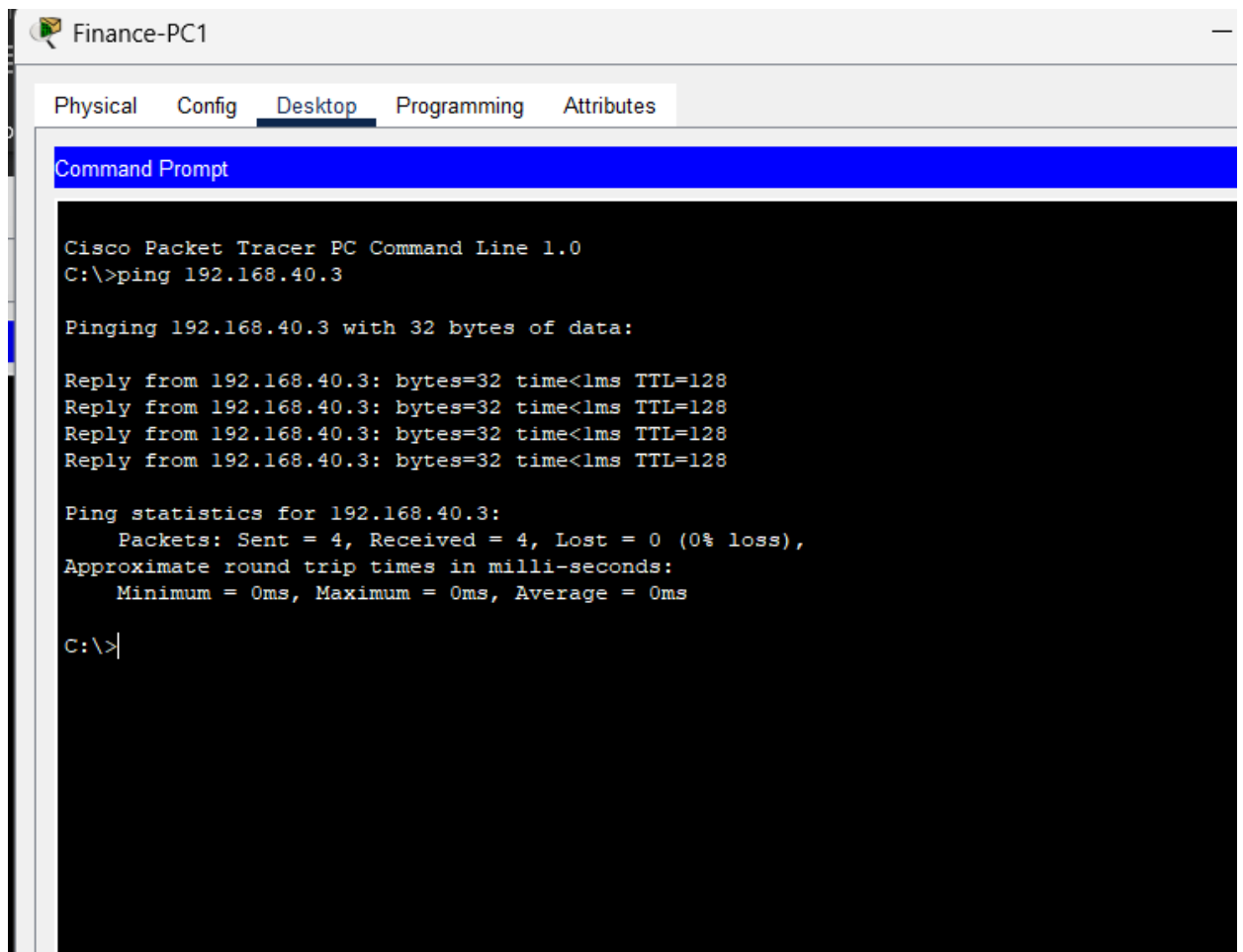
e. HR dept



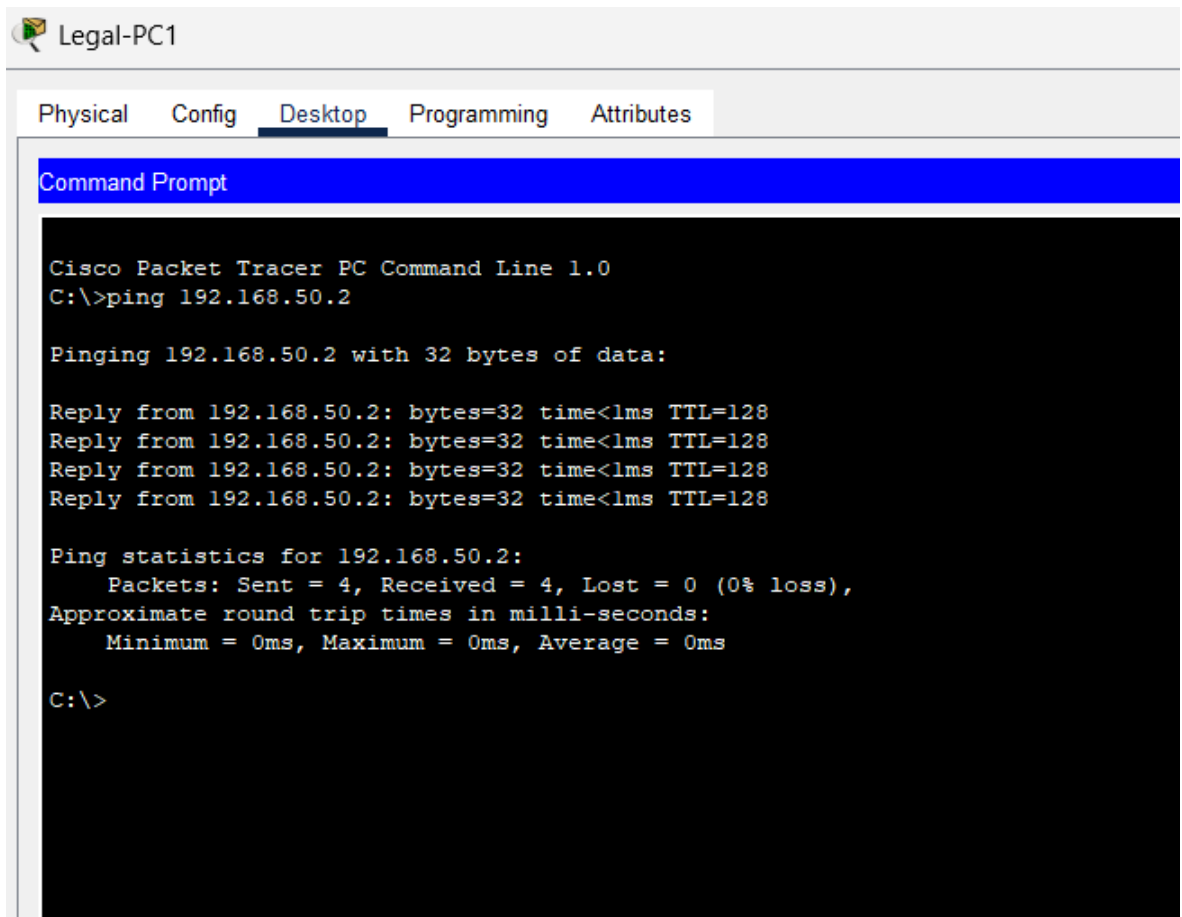
f. IT dept



g. Finance dept



h. Legal dept



With the use of the multilayers switch communication between the inter-VLAN should be enabled (ip routing) .

- i. This diagram below describes the vlans on the multicore

```
Core-MLS#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Admin_Department	active	
20	HR_Department	active	
30	IT_Department	active	
40	Finance_Department	active	
50	Legal_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- j. The diagram below depicts the trunk interfaces found on the core

```
Core-MLS#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1
Fa0/5     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40,50
Fa0/2     1,10,20,30,40,50
Fa0/3     1,10,20,30,40,50
Fa0/4     1,10,20,30,40,50
Fa0/5     1,10,20,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40,50
Fa0/2     1,10,20,30,40,50
Fa0/3     1,10,20,30,40,50
Fa0/4     1,10,20,30,40,50
Fa0/5     1,10,20,30,40,50
```

k. The details of the the ip route of the switch

```
Core-MLS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50
```

l. Details of the ip interfaces found on the core switch all the needed interfaces are UP together with their Protocols.

```
Core-MLS#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	up	up
FastEthernet0/4	unassigned	YES	unset	up	up
FastEthernet0/5	unassigned	YES	unset	up	up
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	down	down
GigabitEthernet0/2	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.1	YES	manual	up	up
Vlan20	192.168.20.1	YES	manual	up	up
Vlan30	192.168.30.1	YES	manual	up	up
Vlan40	192.168.40.1	YES	manual	up	up
Vlan50	192.168.50.1	YES	manual	up	up

7. Advantages:

- Improved performance and less congestion due to the VLAN segmentation, limiting broadcasting issues.
- Segmentizing the VLANS enhances security because penetration in one department will not easily affect another department.
- Isolating the fault and troubleshooting is much easier since the effect will not be felt by all switches, it will only be the one with the issue.

8. Disadvantages:

- If the multilayered switch fails, communication will only be within the respective departments, inter-VLAN will be paused until resolved.
- It requires that every switch be connected to the core switch in order for it to have communication with other departments, leading to a lot of cable use which could be complex as more are added.

9. Conclusion:

It can now be concluded that star topology was very efficient and that it is a reliable network design for connecting various and significant number of devices to it. The structure of the star clearly shows that it will be easier to identify any point of failure because all the departments operate at different VLANS. The core switch acts as a “hub” since it distributes the information across the entire system and all the devices are connected to it, however, this may pose a risk because should the core fail the entire setup is at risk of losing communication. Therefore, star topology not only brings simplicity, but it also offers security and reliability in this case making it quite suitable for modern systems where security is a big risk. However, this does not make it free of risks, it’s only about managing it effectively. The star can now be said to be the most effective and suitable for a variety of organizations.